

System 2, the cogitative one, is what we normally call *thinking*. It is what you use in a classroom, as it is effortful (even for Frenchmen), reasoned, slow, logical, serial, progressive, and self-aware (you can follow the steps in your reasoning). It makes fewer mistakes than the experiential system, and, since you know how you derived your result, you can retrace your steps and correct them in an adaptive manner.

Most of our mistakes in reasoning come from using System 1 when we are in fact thinking that we are using System 2. How? Since we react without thinking and introspection, the main property of System 1 is our lack of awareness of using it!

Recall the round-trip error, our tendency to confuse “no evidence of Black Swans” with “evidence of no Black Swans;” it shows System 1 at work. You have to make an effort (System 2) to override your first reaction. Clearly Mother Nature makes you use the fast System 1 to get out of trouble, so that you do not sit down and cogitate whether there is truly a tiger attacking you or if it is an optical illusion. You run immediately, before you become “conscious” of the presence of the tiger.

Emotions are assumed to be the weapon System 1 uses to direct us and force us to act quickly. It mediates risk avoidance far more effectively than our cognitive system. Indeed, neurobiologists who have studied the emotional system show how it often reacts to the presence of danger long before we are consciously aware of it—we experience fear and start reacting a few milliseconds before we realize that we are facing a snake.

Much of the trouble with human nature resides in our inability to use much of System 2, or to use it in a prolonged way without having to take a long beach vacation. In addition, we often just forget to use it.

Beware the Brain

Note that neurobiologists make, roughly, a similar distinction to that between System 1 and System 2, except that they operate along anatomical lines. Their distinction differentiates between parts of the brain, the *cortical* part, which we are supposed to use for thinking, and which distinguishes us from other animals, and the fast-reacting *limbic* brain, which is the center of emotions, and which we share with other mammals.

As a skeptical empiricist, I do not want to be the turkey, so I do not want to focus solely on specific organs in the brain, since we do not observe brain functions very well. Some people try to identify what are called the neural correlates of, say, decision making, or more aggressively the neural “substrates” of, say, memory. The brain might be more complicated